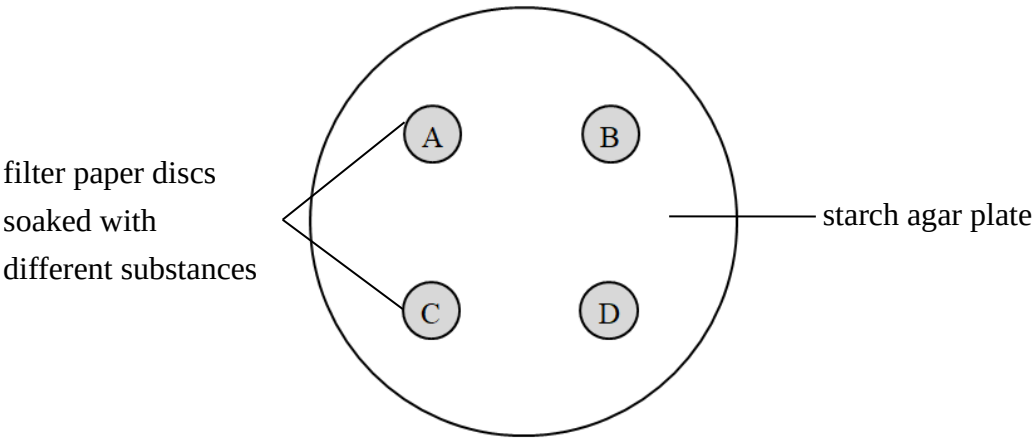
Structured questions

|!|BCELA0070000001|!| ★★ **SI**

An investigation was carried out to study the effects of a juice from the mouth cavity and an extract from germinating mung beans on a starch agar plate. Filter paper discs of the same size were prepared and each disc had been soaked in a different solution for 5 minutes according to the following table.

Filter paper disc	Soaked in
A	juice X from the mouth cavity
B	juice X from the mouth cavity + pH 2 buffer solution
C	fresh extract from germinating mung beans
D	boiled extract from germinating mung beans

After soaking, the filter paper discs were placed on a starch agar plate as shown below. The plate was then incubated for 2 hours. Afterwards, the discs were removed and the plate was dipped into iodine solution.



(a) State and explain the colour observed in spots A and B after the plate was dipped into iodine solution. (4 marks)

(b) Suggest an explanation for each of the following observations:

(i) The colour resulted in spot C is similar to that in spot A. (2 marks)

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(ii) The colour resulted in spot D is similar to that in spot B. (2 marks)

_____

_____

##

(a) Spot A would turn pale brown. (1)
Starch in spot A has been broken down by amylase in juice X. (1)
Spot B would turn blue black. (1)
Starch would still be present because the low pH would have stopped the amylase in juice X from working properly. (1)

(b) (i) The extract from germinating mung beans contains amylase (1)
which catalyses the breakdown of starch to maltose. (1)
Therefore, spot C would turn pale brown after contact with iodine solution, indicating the absence of starch.

(ii) The extract from germinating mung beans contains the enzyme amylase.
Boiling has caused the enzyme to become denatured. (1)
The enzyme could not catalyse the breakdown of starch. (1)
Therefore, spot D would turn blue black after contact with iodine solution, indicating the presence of starch.

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Analysing data	1
	Applying concepts	1

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|!|BCELA0070000002|!| ★★ SI

Digestive juice P was collected from the gall bladder, and digestive juice Q was collected from the pancreas. These juices were mixed in different combinations with equal volumes of peanut oil. The results are shown below.

Tube	Contents	Result
A	Juice P and peanut oil	Oil droplets formed.
B	Juice Q and peanut oil	A small quantity of fatty acid is detected.
C	Juice P, juice Q and peanut oil	A larger quantity of fatty acid is detected.
D	Boiled juice P and peanut oil	?
E	Juice P, boiled juice Q and peanut oil	Oil droplets formed.

(a) Name the digestive juices P and Q. (2 marks)

(b) Explain the results of tubes A and B. (2 marks)

(c) Explain the difference between the results of tubes B and C. (1 mark)

(d) Deduce and explain the results of tube D. (3 marks)

##

(a) P is bile. (1)
Q is pancreatic juice. (1)

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(b) In tube A, juice P causes the emulsification of oil, forming oil droplets. (1)
In tube B, juice Q contains an enzyme which causes the breakdown of lipids to fatty acids and glycerol. (1)

(c) Juice P in tube C emulsifies oil to form oil droplets, providing a larger surface area for the enzyme in juice Q to act on. (1)

(d) Oil droplets would be observed in tube D, similar to that of tube A. (1)
Juice P contains bile salts / does not contain enzymes. (1)
Its effect on oil is not affected by heating. (1)

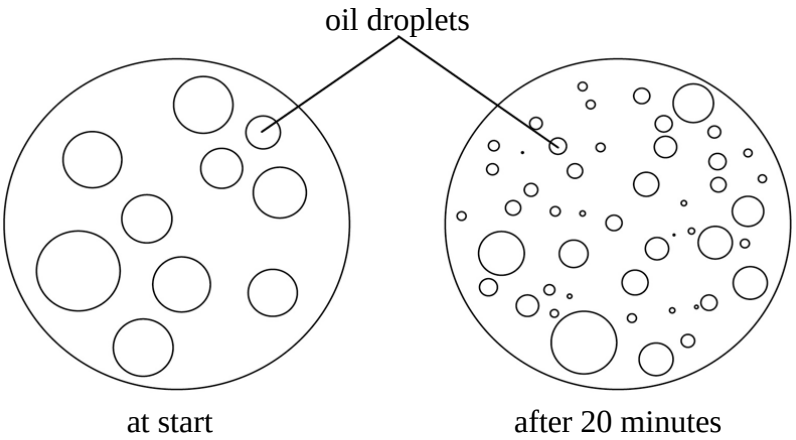
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	1
(d)	Analysing data	1
	Applying concepts	2

|!|BCELB0070000003|!| ★ **SI**

A digestive juice X collected from a mammal was added to a mixture of oil and water. At the beginning, the mixture was examined under a microscope at 200x and then examined again after 20 minutes. The appearance of the two mixtures is shown below.



(a) Describe the effect of digestive juice X on oil. (1 mark)

(b) Based on your answer to (a), how does digestive juice X help the digestion of oil in the human digestive system? (2 marks)

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(c) Name juice X and state its origin of production. (2 marks)

(d) May proposed that digestive juice X contains an enzyme for the action above. Suggest an experiment to prove whether the action above is caused by enzyme in juice X or not. (3 marks)

##

(a) Juice X causes the emulsification of oil, forming oil droplets. (1)

(b) Juice X physically breaks lipids into many small oil droplets, providing a large surface area for lipase to act on. (1)

The rate of lipid digestion is faster. (1)

(c) Juice X is bile. (1)

It is produced by the liver. (1)

(d) Boil juice X before it is mixed with oil and water. (1)

If emulsification of oil cannot be observed, this shows that it is caused by enzyme which is denatured by heating. (1)

If emulsification of oil could be observed, this shows that emulsification is not caused by enzyme in juice X. (1)

(8 marks)

##

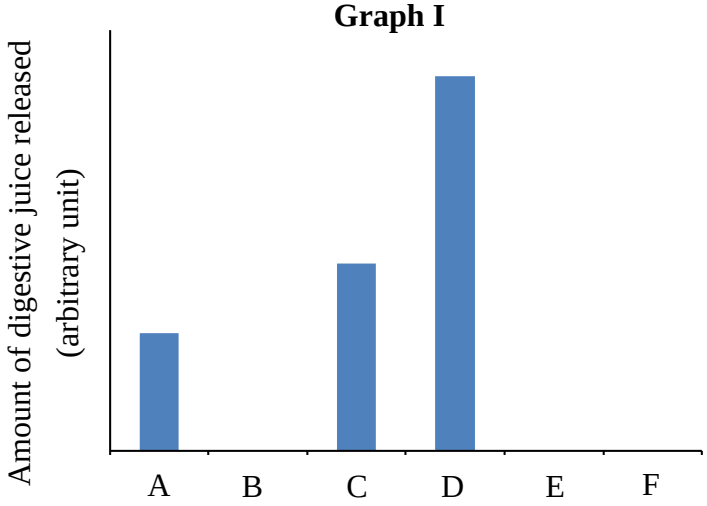
	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Understanding basic concepts	2
(d)	Practical skills	3

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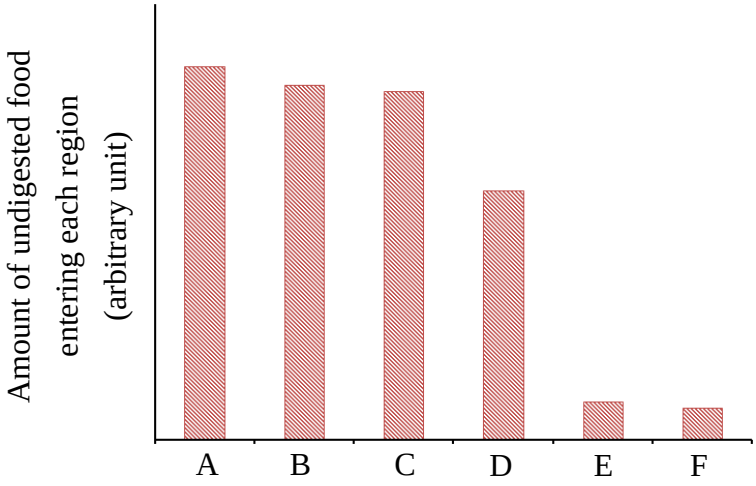
|!|BCELA0070000004|!| ★★

Michael analysed the amount of undigested food and the amount of digestive juice released in six regions (A to F) along the alimentary canal of rats. The graphs below show the results.

Graph I



Graph II



(a) From the graphs above, deduce which region of the alimentary canal is the site where most of digestion occurs. (3 marks)

(b) Why does the amount of undigested food decrease only slightly from regions A to B even though region A has secreted digestive juice? (2 marks)

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(c) Describe the physical and chemical digestion in region C. (3 marks)

(d) What are regions E and F? Explain your answer with reference to the graphs. (3 marks)

##

(a) Region D (1)
The amount of digestive juice released is the greatest in this region. (1)
The amount of undigested food dropped the most drastically in this region. (1)

(b) Region A releases only one type of digestive enzyme amylase, therefore only carbohydrates in the food can be digested. (1)
The duration for the food to stay in the regions A and B is short. (1)

(c) The food in region C is torn into smaller pieces by its churning action. (1)
Churning also helps mix the food with gastric juice. (1)
Gastric juice contains proteases to catalyse the breakdown of proteins into peptides. (1)

(d) Region E is the large intestine. (1)
Region F is the anus. (1)
These two regions do not release digestive juice. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	2
(c)	Understanding basic concepts	3
(d)	Analysing data	1
	Applying concepts	2

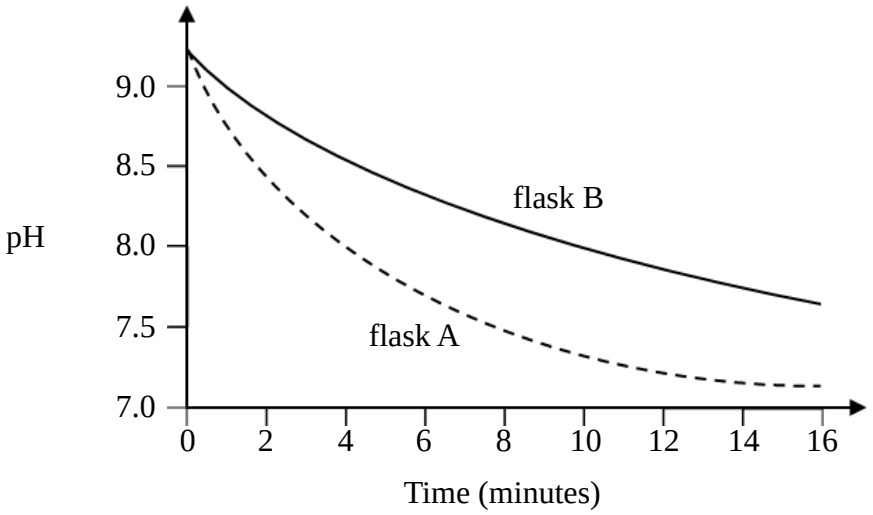
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|!|BCELB0070000005|!| ★ **SI**

An experiment was carried out to investigate the effect of bile salts on the digestion of lipids. Two flasks were set up:

- Flask A contains 10 cm^3 of whole milk, 2 g of sodium carbonate, 2 cm^3 of lipase solution and 2 cm^3 of bile salt solution.
- Flask B contains 10 cm^3 of whole milk, 2 g of sodium carbonate, 2 cm^3 of lipase solution and 2 cm^3 of distilled water.

The pH of each reaction mixture was recorded over 16 minutes using a pH meter connected to a computer. The results are shown in the graph below.



(a) What is the purpose of including sodium carbonate in the reaction mixture? (1 mark)

(b) Why did the pH of the reaction mixtures decrease during the experiment? (2 marks)

(c) What evidence from the graph suggests that bile salts help in lipid digestion? (2 marks)

(d) In flask A, the pH of the mixture remained more or less the same after 14 minutes. Suggest ***two*** reasons for this. (2 marks)

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##

(a) Sodium carbonate solution is alkaline. It provides an alkaline medium for lipase. (1)

(b) Lipase catalyses the breakdown of lipids in milk to form glycerol and fatty acids. (1)

Fatty acids are acidic / lower the pH of the mixture. (1)

(c) The pH of the reaction mixture dropped faster in flask A than in flask B. (1)

This shows that fatty acids are formed at a higher rate in flask A. (1)

(d) All the lipids in milk have been broken down. (1)

The pH of the mixture has become unsuitable for the efficient functioning of lipase. (1)

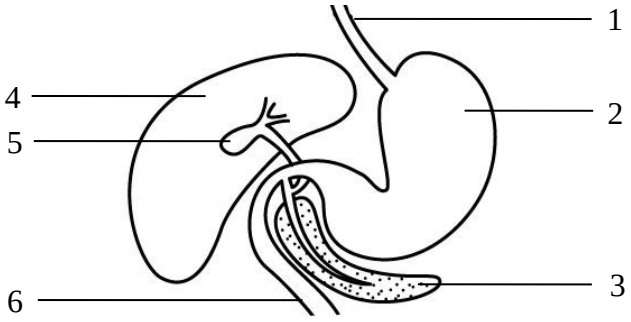
(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Applying concepts	2

|!|BCELB0070000006|!| ★ **Cross topic**

The diagram below shows a part of the human digestive system.



(a) Using the letters in the diagram, state all the parts which produce juices required for the complete digestion of a fried egg. (2 marks)

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(b) How do structures 3 and 4 work together to digest lipids? (5 marks)

(c) How do structures 3 and 4 work together to regulate the blood glucose level? (4 marks)

##

(a) structures 2, 3, 4 and 6 (0.5 mark for each correct answer) (2)

(b) Structure 4 secretes bile which contains bile salts. (1)
Bile salts emulsify lipids to form small oil droplets. (1)
This increases the surface area of lipids for the action of lipase. (1)
Structure 3 secretes pancreatic juice which contains lipase. (1)
Lipase catalyses the breakdown of lipids into glycerol and fatty acids. (1)

(c) When the blood glucose level rises, structure 3 detects the change and it secretes more insulin and less glucagon. (1)
Insulin stimulates structure 4 to convert glucose to glycogen for storage. (1)
When the blood glucose level falls, structure 3 detects the change and it secretes more glucagon and less insulin. (1)
Glucagon stimulates structure 4 to convert the stored glycogen to glucose. (1)

(11 marks)

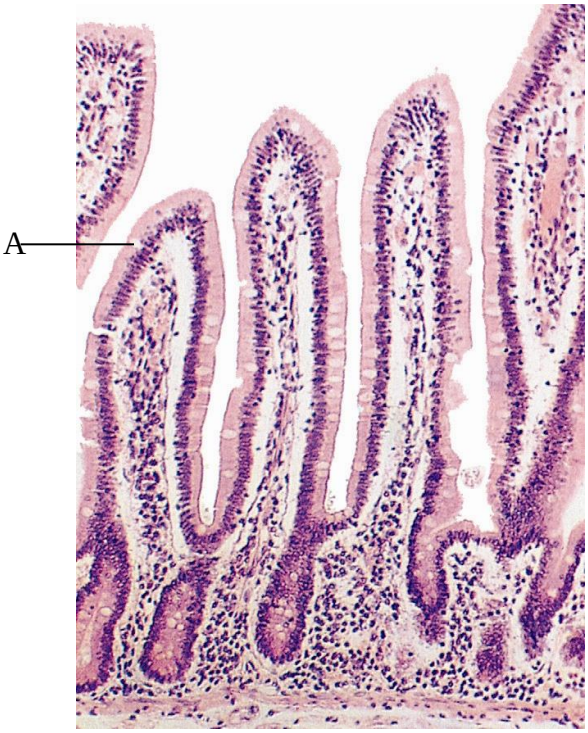
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	5
(c)	Applying concepts	4

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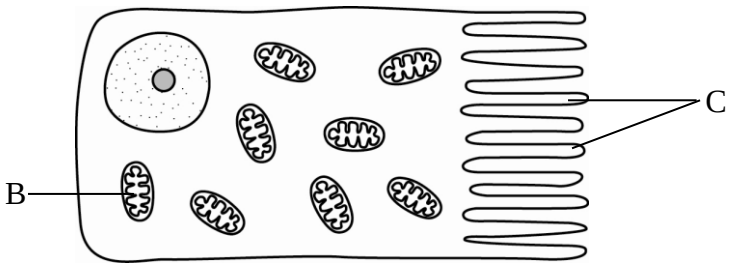
|!|BCELB0070000007|!| ★ **STSE**

The following micrograph shows a longitudinal section of the small intestine.



(a) Referring to ***two*** features shown in the micrograph, explain how the small intestine is adapted for absorption of digested food substances. (4 marks)

(b) The following diagram shows a cell from structure A which is shown in the micrograph.



(i) There are abundant organelles B in this cell. Explain how these organelles facilitate the absorption of digested food substances by the cells. (2 marks)

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(ii) Coeliac disease is a disease of the digestive system. For people suffering from coeliac disease, structures C in this cell are damaged.
People with coeliac disease can digest proteins, but they have a low concentration of amino acid in their blood. Explain why. (2 marks)

##

(a) The intestinal wall has many finger-like projections / villi. (1)
This feature provides a large surface area for absorption. (1)
The epithelium of villi is only one-cell thick. (1)
This shortens the distance of diffusion / transport of digested food substances. (1)

(b) (i) Mitochondrion (Structure B) is the site of respiration / ATP production, (1)
which supplies energy for active transport of products of digestion. (1)

(ii) The surface area of structure C is reduced (1)
and so there is less absorption of amino acids. (1)
OR
The carrier proteins in structure C are damaged (1)
and so there is less absorption of amino acids. (1)

(8 marks)

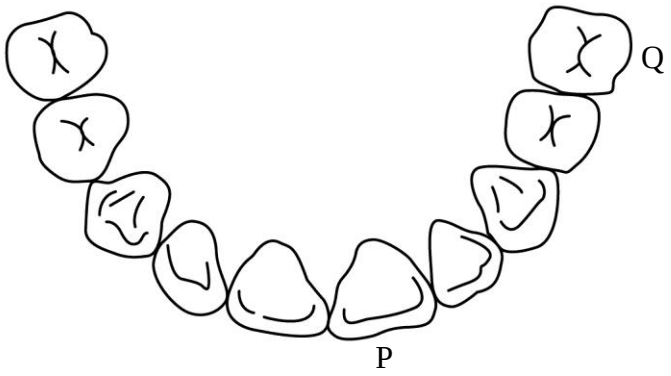
##

	Skills assessed	Marks
(a)	Analysing data	4
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2

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|!|BCELB0070000008|!| ★

The figure below shows the teeth in the lower jaw of a three-year-old boy.



(a) (i) Identify tooth P and tooth Q. (2 marks)

(ii) Describe how tooth P and tooth Q are adapted to their functions. (2 marks)

(b) (i) If the upper jaw contains the same number and types of teeth as that of the lower jaw, state the dental formula of this boy. (1 mark)

(ii) Give ***two*** differences between the teeth set of this boy and that of an adult. (2 marks)

(c) State and explain the importance of ***one*** type of vitamin for building teeth in children. (2 marks)

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##

(a) (i) Tooth P: incisor (1)

Tooth Q: molar (1)

(ii) Tooth P is chisel-shaped which facilitates the cutting of food. (1)

Tooth Q has a broad top with cusps which can crush and grind food. (1)

(b) (i)

$$i\frac{2}{2}\ c\ \frac{1}{1}\ pm\ \frac{0}{0}\ m\ \frac{2}{2}$$ (1)

(ii) Any ***two*** of the following differences: (2)

Teeth set of the boy	Teeth set of the adult
• does not have premolars	• has premolars
• has fewer molars	• has more molars
• the total number of teeth is smaller	• the total number of teeth is greater

(c) vitamin D (1)

It helps the absorption of calcium which is essential for building teeth. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	2
(c)	Understanding basic concepts	2

|!|BCELB0070000009|!| ★ SI

Phenol red solution can be used as pH indicator. The table below shows the colour of phenol red solution at different pH values.

pH	Colour of phenol red solution
< 6.8	yellow
6.8 – 8.2	pink
> 8.2	red

An experiment was performed to study the digestion of lipids. Milk, lipase solution and phenol red solution were mixed in a test tube. The mixture changed colour after 10 minutes.

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(a) Describe and explain the colour change of the mixture. (3 marks)

(b) Milk, lipase solution, phenol red solution and bile are to be mixed in another test tube. Predict whether the same results would be observed. (2 marks)

(c) Which substance in bile leads to the results in (b)? Describe the role of this substance in the digestion of lipids. (3 marks)

##

(a) The mixture changed to yellow. (1)
This is because lipase catalysed the breakdown of lipids into fatty acids and glycerol. (1)
The fatty acids lowered the pH of the mixture. (1)
(b) The mixture would change to yellow, (1)
but within a shorter period of time. (1)
(c) Bile salts (1)
Bile salts emulsify lipids into small droplets. (1)
This increases the surface area for the enzyme lipase to act on. (1)

(8 marks)

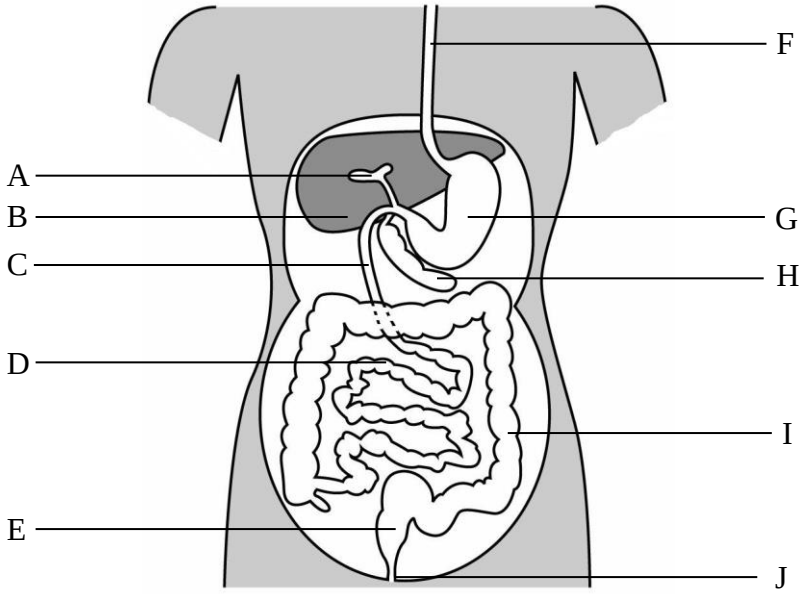
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	2
(c)	Understanding basic concepts	3

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|!|BCELC0070000010|!|

The diagram below shows the human digestive system with structures labelled as A to J.



(a) Which labelled structures of the digestive system produce digestive juices that are alkaline? (2 marks)

(b) In which labelled structure can protease and lipase first be found along the alimentary canal? (2 marks)

(c) Structure A contains a digestive juice.

(i) What is the digestive juice in structure A? (1 mark)

(ii) Explain why this digestive juice is important in digestion. (2 marks)

(d) Name ***two*** labelled structures in the diagram that have functions other than digestion or absorption. State their functions. (4 marks)

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##

(a) Structures B, C, D and H (2)

(b) Protease is first found in structure G. (1)

Lipase is first found in structure C. (1)

(c) (i) The digestive juice is bile. (1)

(ii) It contains bile salts which emulsify lipid into small droplets. (1)

This increases the surface area of lipid for lipase to act on. (1)

OR

It contains sodium hydrogencarbonate, which is alkaline. (1)

It neutralizes the acidic chyme from the stomach and provides an optimum pH for lipase to function. (1)

(d) Any ***two sets*** of the following: (4)

Organ	**Function**
Stomach (1)	body defence (some pathogens are killed by the gastric juice) (1)
Liver (1)	regulation of blood glucose level / deamination / storage of iron and fat-soluble vitamins (1)
Pancreas (1)	secretion of insulin and glucagon to regulate the blood glucose level (1)

(11 marks)

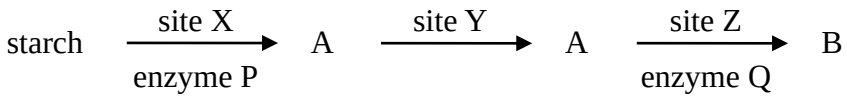
##

	Skills assessed	**Marks**
(a)	Analysing data	2
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2
(d)	Analysing data	2
	Understanding basic concepts	2

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|!|BCELB0070000011|!| ★

The flowchart below summarizes the processes in the digestion of carbohydrates. Sites X, Y and Z are certain parts of the alimentary canal where digestion occurs. A and B are the products of enzymatic reactions.



(a) Identify site X and enzyme P. (2 marks)

(b) What is A? (1 mark)

(c) After molecule A and enzyme P entered site Y, enzyme P has become denatured. Explain how this happens. (2 marks)

(d) Name site Z. Describe how molecule A is digested in site Z. (3 marks)

(e) How is molecule B absorbed by the body? (2 marks)

##

(a) Site X is the mouth cavity. (1)
Enzyme P is salivary amylase. (1)
(b) maltose (1)
(c) Site Y (stomach) secretes gastric juice which contains hydrochloric acid. (1)
The acidic gastric juice causes the denaturation of enzyme P. (1)
(d) Site Z is the small intestine. (1)

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The epithelial cells in the wall of the small intestine contain carbohydrases (e.g. maltase). (1)

Maltase catalyses the digestion of molecule A (maltose) into glucose. (1)

(e) Molecule B (glucose) is absorbed into the bloodstream (1)

by diffusion and active transport. (1)

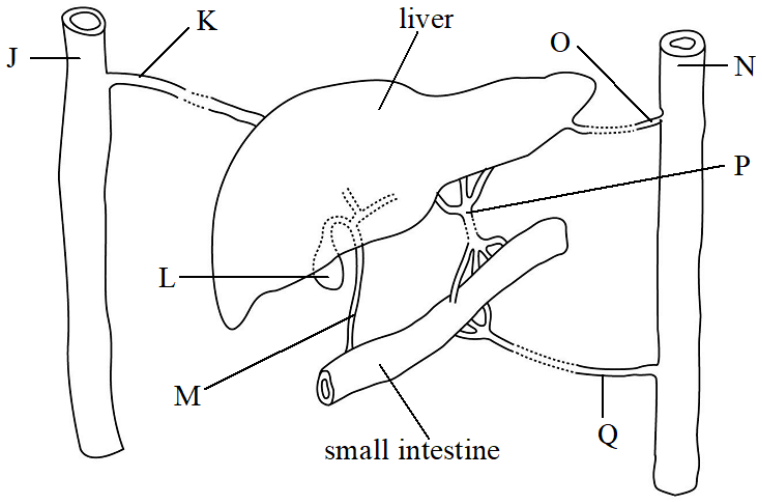
(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	3
(e)	Understanding basic concepts	2

|!|BCELB0070000012|!| ★

The diagram below shows the liver and the associated blood vessels.



(a) What juice is contained in structure L? Describe its effect on cooking oil. (2 marks)

(b) Identify vessels K and P. (2 marks)

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(c) When a person gets starved, which vessel contains blood with the highest concentration of glucose? Explain in relation to the hormone(s) involved. (5 marks)

(d) Blood flows from N to J through the liver in two pathways. Construct flowcharts to show these two pathways. (2 marks)

##

(a) Structure L contains bile. (1)
Bile salts in bile emulsify lipids into smaller droplets. (1)

(b) K is the hepatic vein. (1)
P is the hepatic portal vein. (1)

(c) K (1)
During starvation, the blood glucose level is low. (1)
Less insulin and more glucagon is secreted from the pancreas. (1)
So less glucose is converted into glycogen for storage in the liver. (1)
Meanwhile, glycogen is converted into glucose by the liver and released into K. (1)
Thus, K contains blood with the highest concentration of glucose.

(d) N → O → liver → K → J (1)
N → Q → small intestine → P → liver → K → J (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Analysing data	2
(c)	Analysing data	1
	Applying concepts	4
(d)	Understanding basic concepts	2

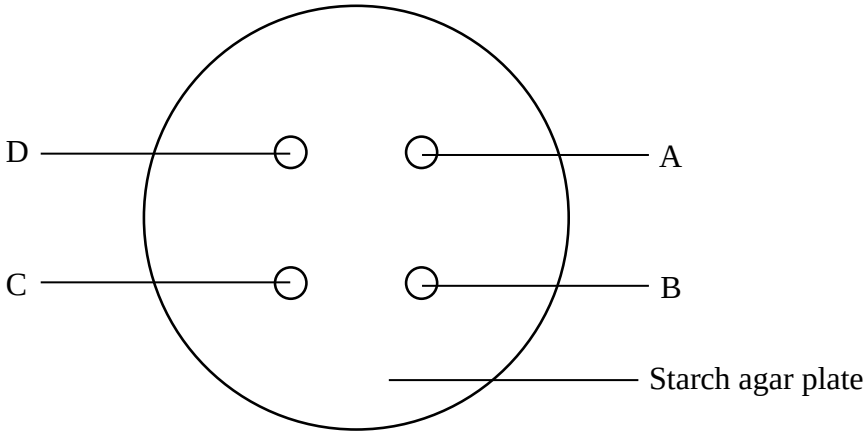
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|!|BCELB0070000013|!| ★ **SI**

In an experiment, Ann first prepared four samples, as shown in the table below.

Sample	Content
A	extract from the gut of a grasshopper
B	human saliva
C	extract from a germinating maize grain
D	extract from the colon of a rat

Ann put the samples into the cavities in a starch agar plate as shown in the diagram below. After five hours, she poured iodine solution onto the starch agar plate. Clear zones are observed around three of the cavities. Only one cavity does not have any clear zone around it.



(a) What does the presence of a clear zone indicate? (1 mark)

(b) Using the letters given, state which cavity does not have a clear zone around it. (1 mark)

(c) Which substance is present in the samples that lead to the formation of the clear zones? (1 mark)

(d) With reference to one of the sample, explain the importance of the presence of the substance answered in (c) to the organisms. (2 marks)

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(e) State and explain the difference in the results if Ann had boiled the human saliva sample at the beginning of the experiment. (2 marks)

##

(a) A clear zone indicates the absence of starch in the agar. (1)

(b) D (1)

(c) amylase (1)

(d) Any ***one set*** of the following: (2)

- The gut of a grasshopper contains amylase which catalyses the digestion of starch in leafy food into maltose. (1)
  Maltose is then further digested into glucose for respiration to release energy. (1)
- Human saliva contains amylase which catalyses the digestion of starch in food into maltose. (1)
  Maltose is then further digested into glucose for respiration to release energy. (1)
- A germinating maize grain contains amylase which catalyses the digestion of starch in the grain into maltose. (1)
  Maltose is then further digested into glucose for respiration to release energy for the growth of the seedling. (1)

(e) There would be no clear zone around the cavity containing sample B (1)
because amylase has been denatured by the high temperature. (1)

(7 marks)

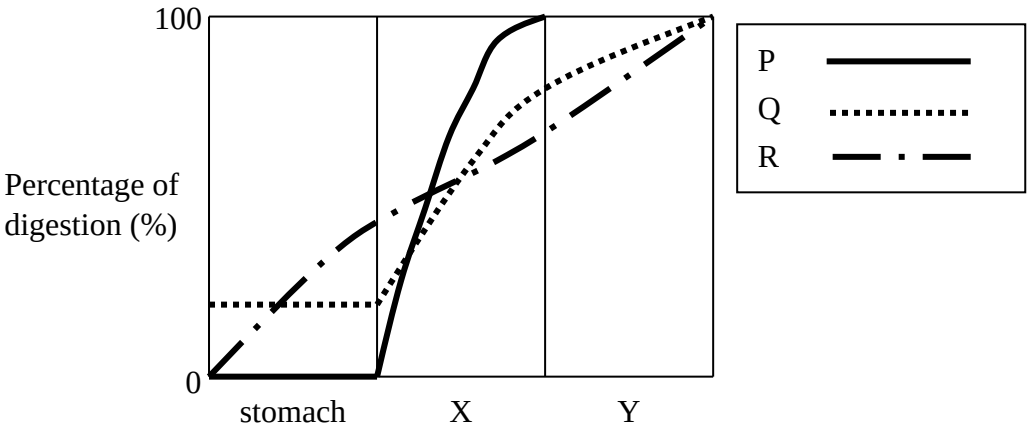
##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
(c)	Understanding basic concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	2

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|!|BCELB0070000014|!| ★

The graph below shows the percentage of digestion of carbohydrate, protein and lipid along different regions of the alimentary canal of a man.



(a) Identify regions X and Y. (2 marks)

(b) Which food substances do P, Q and R represent respectively? (3 marks)

(c) Describe and explain the shape of curve Q. (6 marks)

##

(a) X: duodenum (1)
Y: ileum (1)

(b) P: lipid (1)
Q: carbohydrate (1)
R: protein (1)

(c) The percentage of digestion of starch is about 20% when food enters the stomach. (1)

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This is because digestion of starch starts in the mouth cavity where salivary amylase catalyses the digestion of starch to maltose. (1)

The percentage of digestion of starch remains unchanged in the stomach. (1)

This is because the acidity in the stomach is too high for salivary amylase to function. (1)

The percentage of digestion of starch increases along region X and finally reaches 100% in region Y. (1)

This is because starch is digested by pancreatic amylase from the pancreatic juice. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	3
(c)	Analysing data	3
	Applying concepts	3

|!|BCELC0070000015|!|

Susan had lunch in a fast food restaurant. The table below shows the nutritional value of her meal.

Foodstuff	Carbohydrate (g)	Lipid (g)	Protein (g)
fried chicken	15	17	18
sausage and egg burger	27	28	19

(a) (i) Which food substance did Susan take in most in this meal? (1 mark)

(ii) Describe how this food substance is digested in the small intestine. (4 marks)

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(b) Susan's daily protein requirement is 45 g. She obtained 11 g from her breakfast. The total amount of protein she obtained from breakfast and lunch has already exceeded the daily amount required. Describe the fate of the excess proteins in her body. (4 marks)

##

(a) (i) lipid (1)

(ii) The bile salts in bile emulsify lipids into smaller droplets, (1)
which have greater surface areas for lipase to act on. (1)
The lipase in the pancreatic juice secreted by the pancreas (1)
catalyses the digestion of lipids into fatty acids and glycerol. (1)

(b) The proteins are digested into amino acids which are absorbed into the bloodstream. (1)
In the liver, the excess amino acids are deaminated. The amino groups of these amino acids are converted to urea. (1)
Urea is excreted in urine by the kidneys. (1)
The remaining parts of the amino acids are converted to carbohydrates or lipids. (1)

(9 marks)

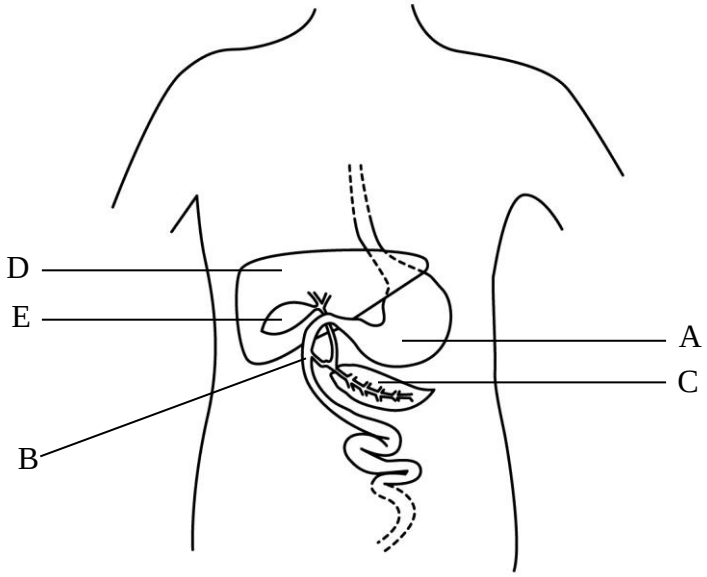
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	4
(b)	Understanding basic concepts	4

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|!|BCELC0070000016|!|

The diagram below shows a certain part of the human digestive system.



(a) (i) Compare the pH values of chyme in A and B. (2 marks)

(ii) Explain the importance of the difference to digestion. (2 marks)

(b) Using the letters in the diagram, list all the parts which produce juices required for the complete digestion of

(i) egg white. (1 mark)

(ii) lipid. (1 mark)

(c) Describe how lipid in chyme is digested physically and chemically in region B. (3 marks)

##

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(a) (i) The chyme in A is acidic, (1)
while the chyme in B is alkaline. (1)
(ii) The low pH in A is optimal for pepsin / protease to work. (1)
The higher pH in B is optimal for amylase and lipase to work. (1)
(b) (i) A and C (1)
(ii) C and D (1)
(c) Peristalsis of the small intestine helps the mixing of chyme with bile and pancreatic juice in region B. (1)
Bile salts in bile emulsify lipids into small droplets, providing a greater surface area for enzyme action. (1)
Lipase in pancreatic juice catalyses the breakdown of lipid to glycerol and fatty acids. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)	Applying concepts	3

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